

Chicago City Council Reform 2026 — Simulation Configurations

This document describes the simulation configurations that we've set up in the `configs/` directory.

Background

Chicago currently elects its City Council from **50 single-member wards** by plurality. This project updates the Metric Geometry and Gerrymandering Group's 2019 city council reform proposals, comparing the status quo against multi-member districts elected by **Single Transferable Vote (STV)**, along with variations on turnout and voter-bloc structure. Representation is measured by simulating elections over an ensemble of districting plans and tallying the seats won by each group's candidates.

Data

- **Geodata:** `./data/chicago_precincts_vap_cvap.gpkg` (1,291 Chicago precincts).
- **Census Data:** Demographic data pulled from the 2020 Decennial Census at the block level and aggregated to precincts.

Configurations

Run Name	Config File	Districts	Seats	Voting Method	Voter Blocs	Turnout (bloc)
Basic – 50×1 Plurality	<code>basic.json</code>	50 × 1	50	Plurality	W-A, B, H	1.00 / 1.00 / 1.00
Low POC Turnout	<code>low-poc-turnout.json</code>	10 × 5	50	FastSTV (5 seats)	W-A, B, H	0.75 / 0.50 / 0.50
Asian Bloc Separate – Basic	<code>asian-seperate-bloc.json</code>	10 × 5	50	FastSTV (5 seats)	A, W, B, H	1.00 / 1.00 / 1.00 / 1.00
10 × 3 STV	<code>10x3-stv.json</code>	10 × 3	30	FastSTV (3 seats)	W-A, B, H	1.00 / 1.00 / 1.00

10 × 5 STV	10x5-stv.json	10 × 5	50	FastSTV (5 seats)	W-A, B, H	1.00 / 1.00 / 1.00
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- **Basic** — roughly the status quo baseline (50 single-member wards, plurality.)
- **Low POC Turnout** — turnout sensitivity: every other run assumes full turnout (1.00) across all blocs, so this run is the only one that varies it. The *W-A* bloc turns out at 0.75, while Black and Latino turnout is lower, at 0.50 each.
- **Asian Bloc Separate** — bloc-structure sensitivity: Asian voters modeled as their own 4th bloc instead of being merged into *W-A*. We are curious to understand how modeling Asian voters as a separate bloc impacts their city council representation if candidates from their preferred slate receive heavier Asian voter support and a modest amount of crossover support from the white voting bloc.
- **10 × 3 STV** and **10 × 5 STV** — the multi-member STV alternatives (3- and 5-seat districts).

Shared parameters

Held constant across every run:

- **Candidate slates:** *W* × 4, *B* × 3, *H* × 3, *A* × 1 (11 total) as a base assumption. However, per-district candidate counts are no longer fixed to this base list: during settings generation, each slate's count is reapportioned in proportion to its VAP share in that specific district (a slate apportioned zero candidates is dropped from that district entirely), then perturbed by `candidate_noise_probability`, which introduces an element of "randomness" into candidate availability at a per-slate-per-district level.
- **Candidate noise:** `candidate_noise_probability`: 0.10 — per slate, per district, a 10% independent chance of gaining a candidate and a 10% chance of losing one - modeling how candidate availability may vary in reality regardless of demographic proportions.
- **Voter models:** `slate_pl` (Plackett-Luce) and `slate_bt` (Bradley-Terry).
- **Focal group:** *A* (Asian).
- **Voters per district:** 10,000 ·
- **Replicates per district:** 10.
- **GerryChain ensemble:** chain length 10,000 · 50 subsampled plans · $\epsilon = 0.05$.
- **Alphas (Dirichlet):** all 1 (uniform within-slate preferences).
- **Tiebreak:** random

Cohesion matrices

Rows are voter blocs, columns are candidate slates; each row sums to 1.

Standard 3-bloc (basic, low-poc-turnout, 10x3-stv, 10x5-stv):

bloc ↓ / slate →	W	B	H	A
W-A	0.46	0.14	0.14	0.26
B	0.15	0.70	0.10	0.05
H	0.15	0.10	0.65	0.10

The W-A row is the VAP-weighted average (81% White / 19% Asian) of the underlying White and Asian rows (below).

Asian Bloc Separate (asian-seperate-bloc):

bloc ↓ / slate →	W	B	H	A
W	0.50	0.15	0.15	0.20
A	0.30	0.10	0.10	0.50
B	0.15	0.70	0.10	0.05
H	0.15	0.10	0.65	0.10

The cohesion parameters here have been selected based on available estimates (from ecological regression) from the earlier 2019 MGGG study on Chicago city council reform, as well as a mayoral election analysis from the University of Illinois-Chicago. The details can be found in the accompanying writeup.

Initial Results

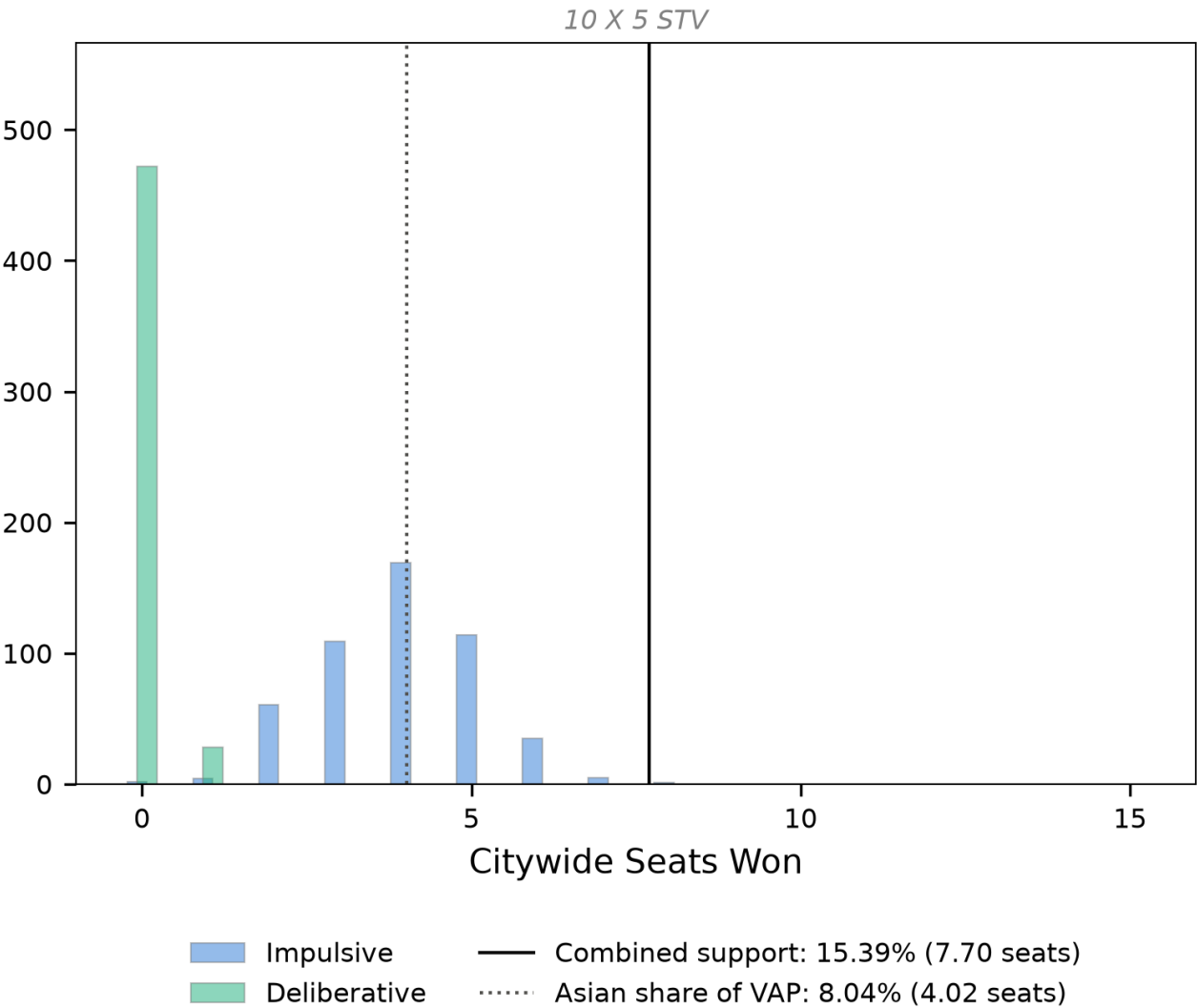
Figures below are from the current pipeline (updated cohesion matrix, full turnout except where noted, and per-district proportional candidate counts with noise). Three figure types are shown for each completed run: a **by-mode histogram** (distribution of citywide seats won by Asian-preferred candidates, one series per voter model, against the Asian share-of-VAP and combined-support reference lines), a **bubble plot** (the same distribution, sized by occurrence count, one row per voter model plus a pooled "Combined" row), and a **by-slate panel** (the same by-mode histogram repeated once per candidate slate — White, Black, Latino, Asian — each against its own share-of-VAP and combined-support reference lines).

Basic – 50×1 Plurality

Basic – 50×1 Plurality hasn't been re-run under the current settings yet, so its figures aren't included here — this section will be updated once that run completes.

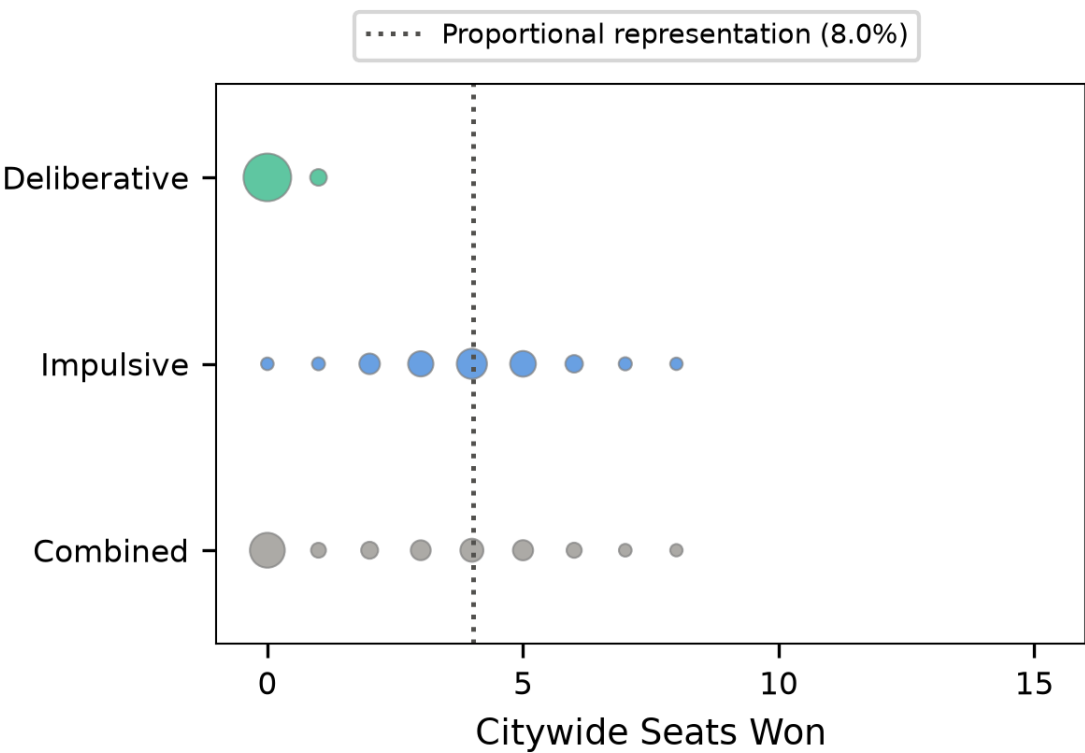
10 × 5 STV

Election Outcomes for Asian-Preferred Candidates



Election Outcomes for Asian-Preferred Candidates

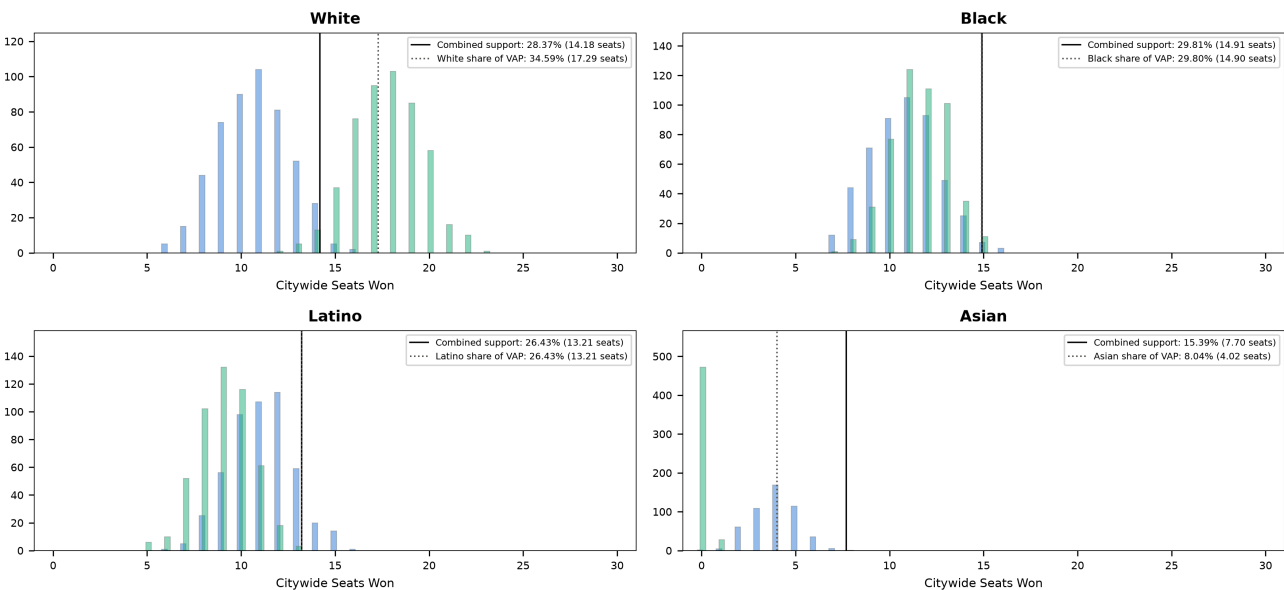
10 X 5 STV



Election Outcomes by Slate

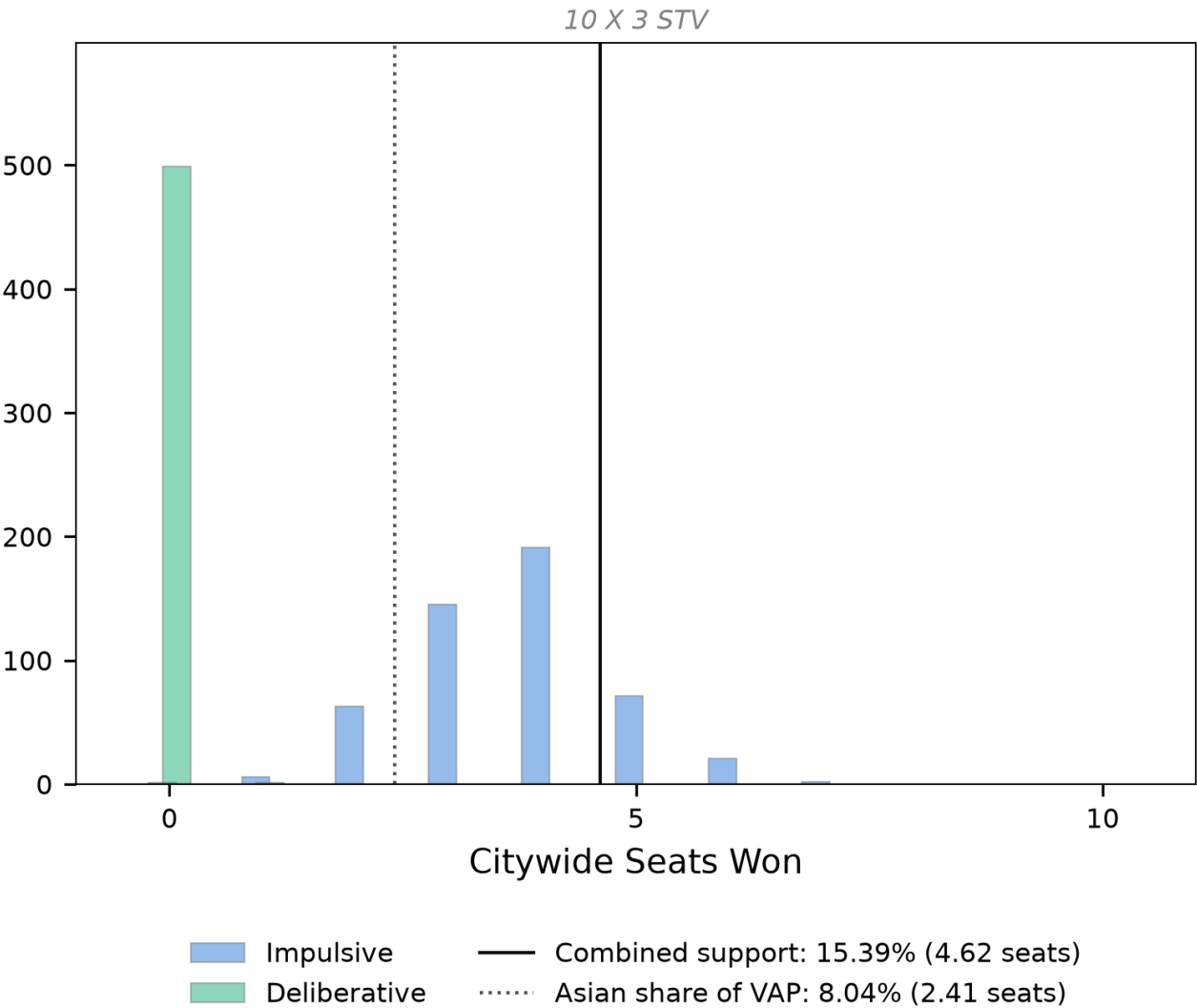
10 X 5 STV

Mode
Impulsive
Deliberative



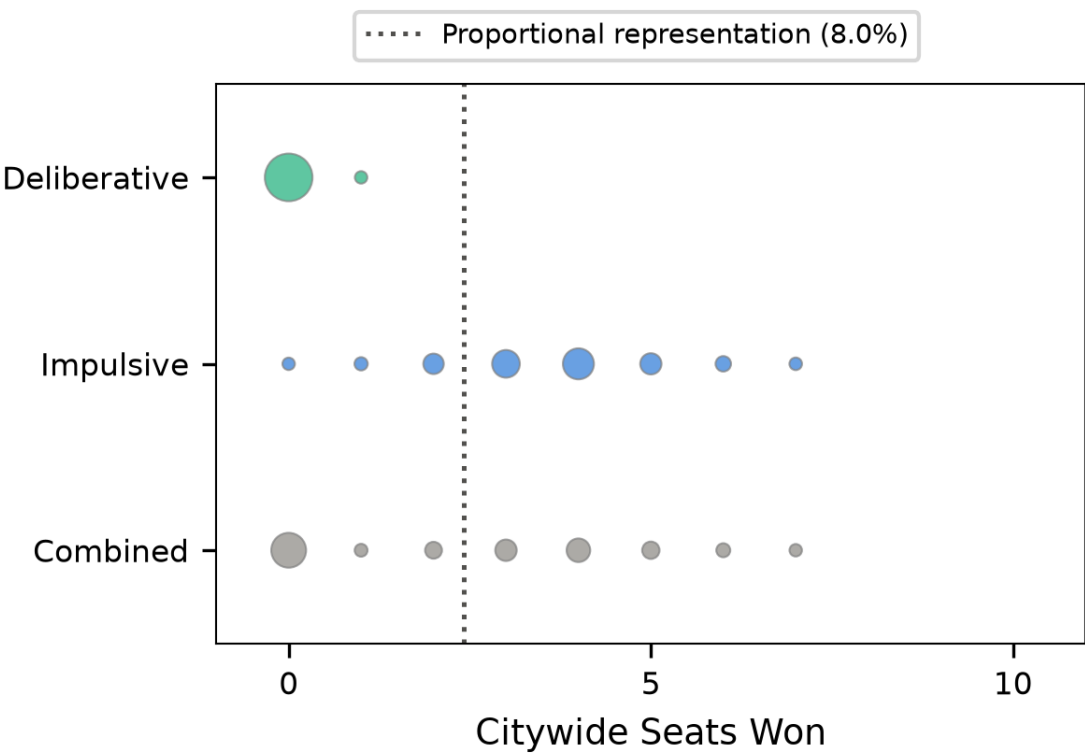
10 × 3 STV

Election Outcomes for Asian-Preferred Candidates



Election Outcomes for Asian-Preferred Candidates

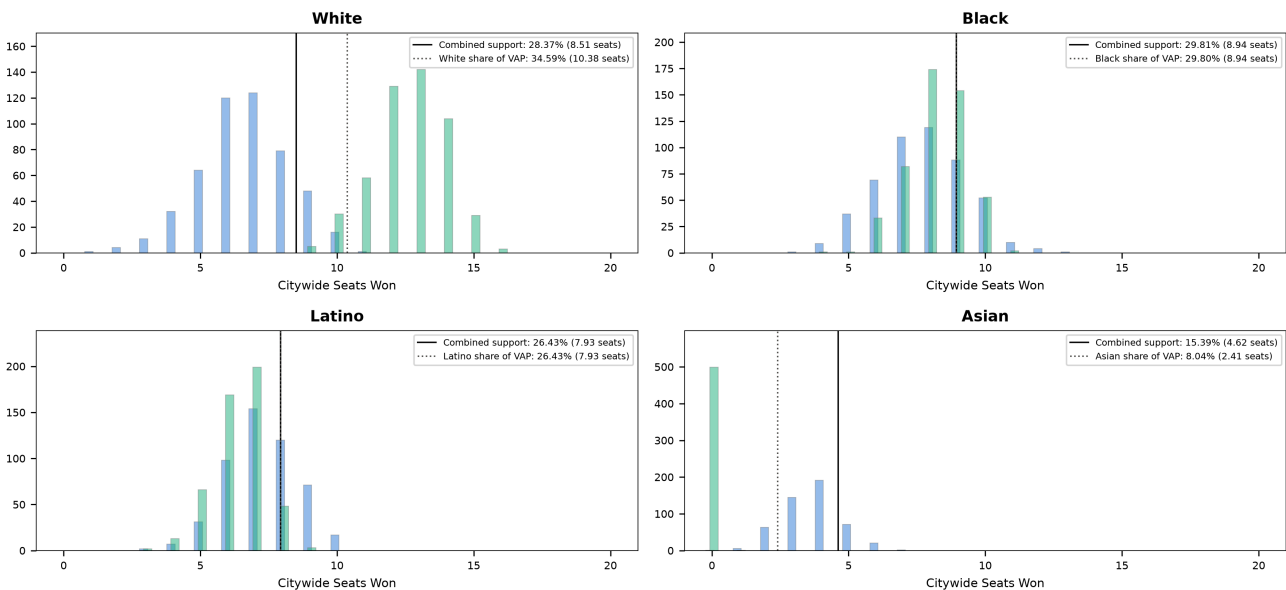
10 X 3 STV



Election Outcomes by Slate

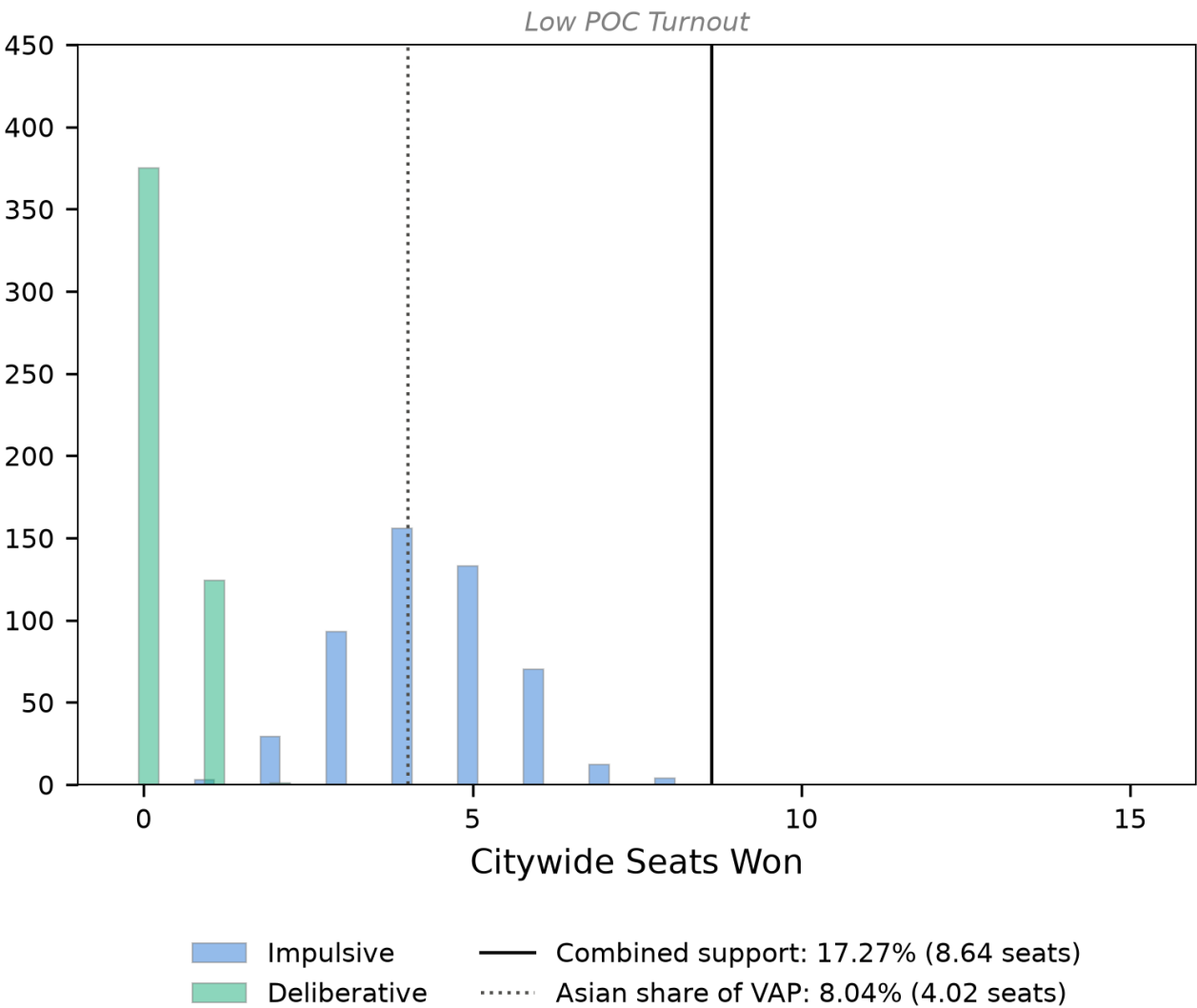
10 X 3 STV

Mode
Impulsive
Deliberative



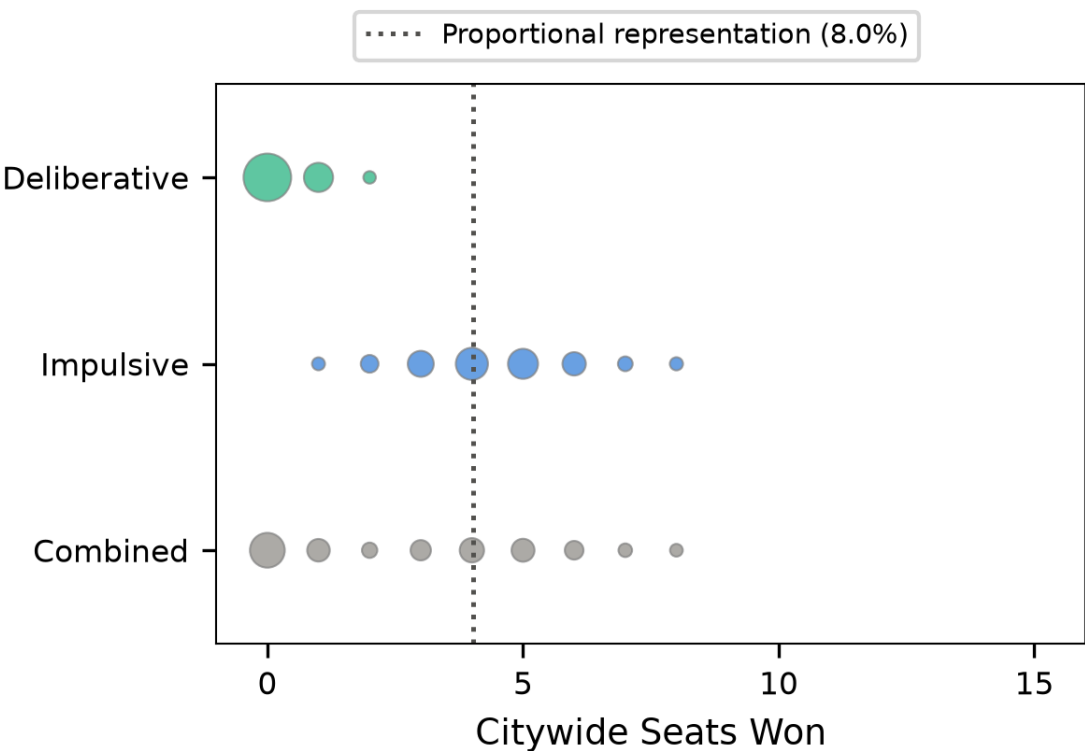
Low POC Turnout

Election Outcomes for Asian-Preferred Candidates



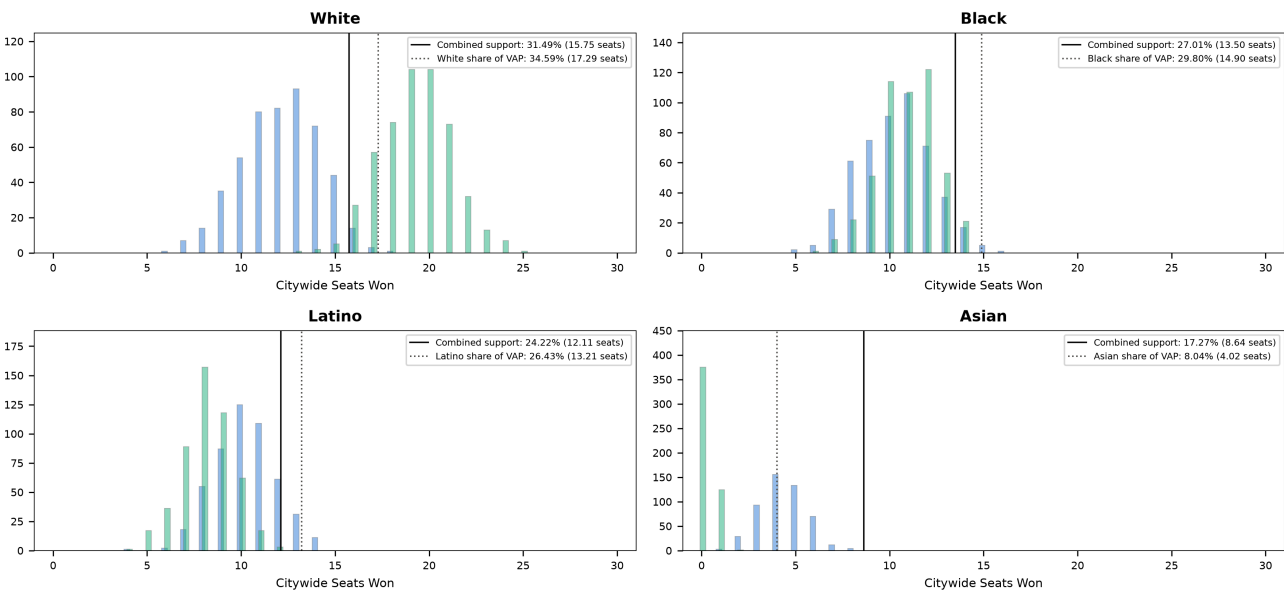
Election Outcomes for Asian-Preferred Candidates

Low POC Turnout

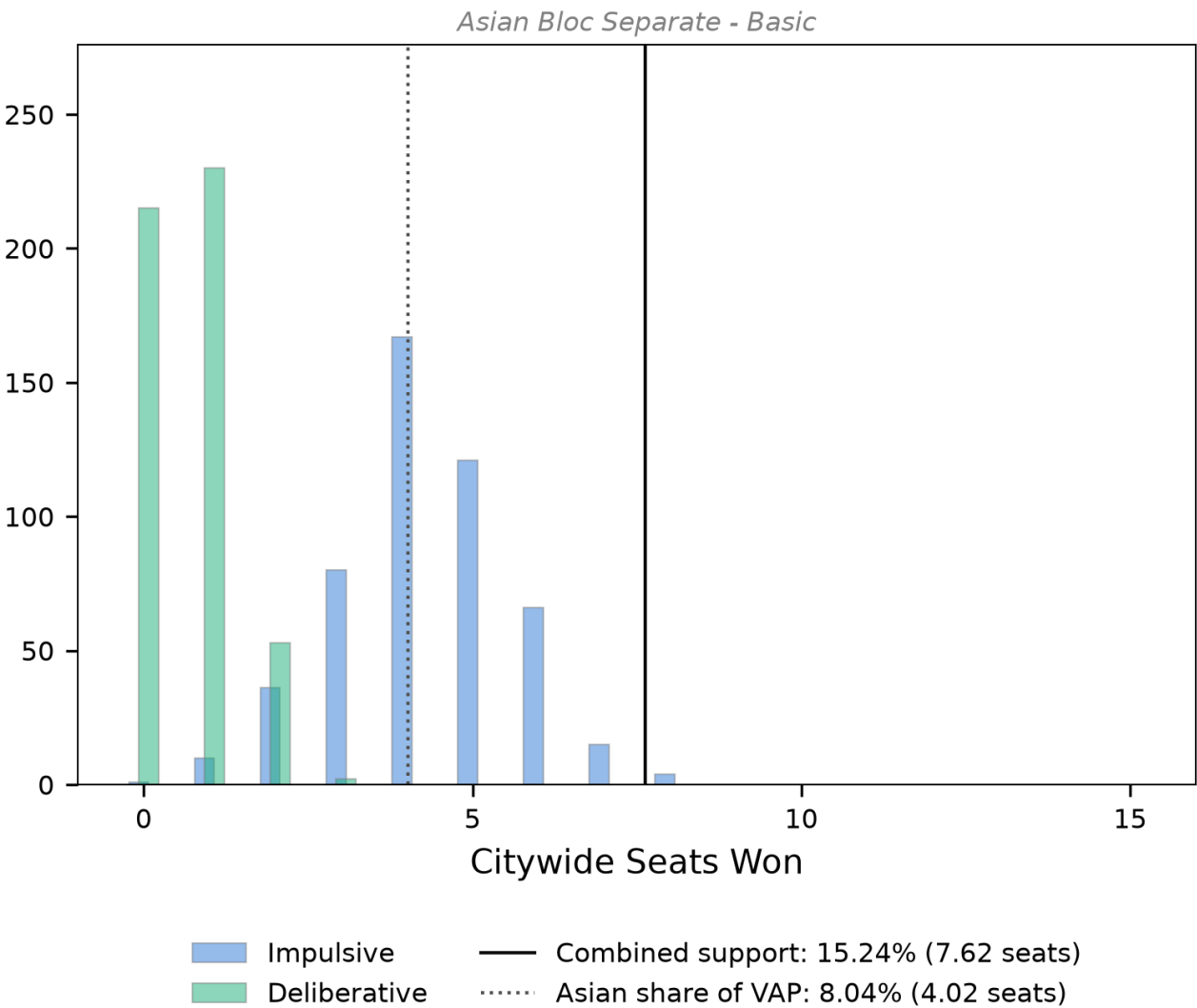


Election Outcomes by Slate
Low POC Turnout

Mode
Impulsive
Deliberative

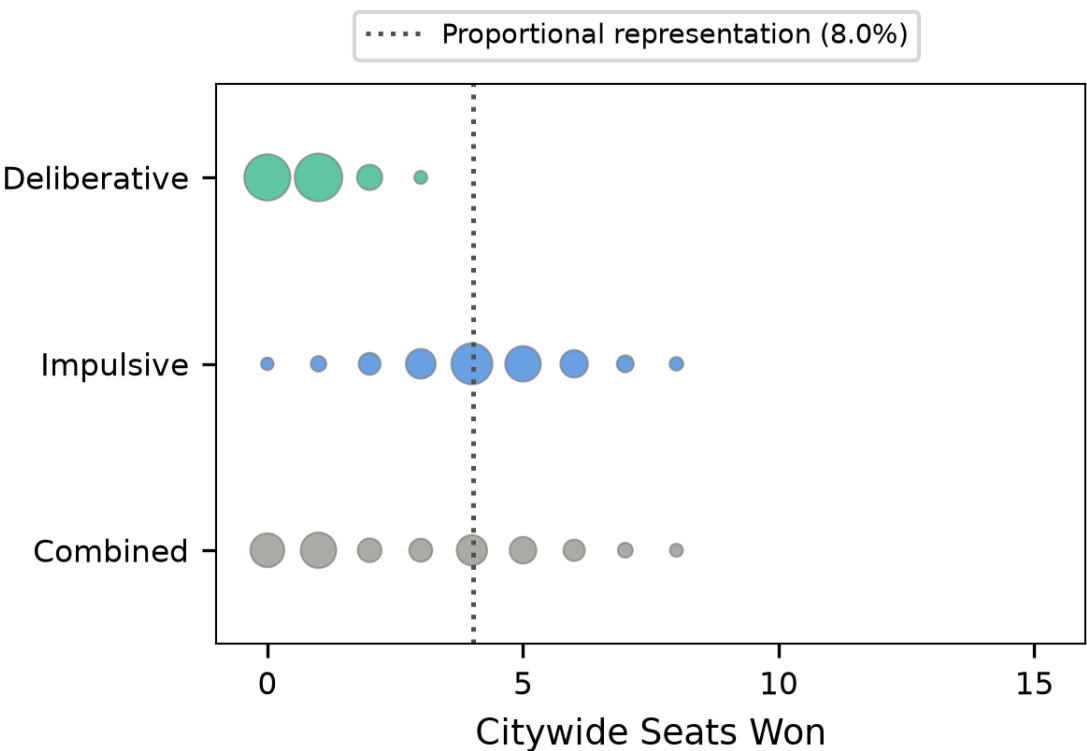


Election Outcomes for Asian-Preferred Candidates



Election Outcomes for Asian-Preferred Candidates

Asian Bloc Separate - Basic



Election Outcomes by Slate
Asian Bloc Separate - Basic

Mode
Impulsive
Deliberative

